# Carlos Jurado

Academic Website — carlosjurado@utexas.edu

## Education

**PhD, Astronomy**; The University of Texas at Austin 2024 -
*Advisor:* Keith Hawkins

**MA, Astronomy**; The University of Texas at Austin 2026
*Thesis:* Dynamical Origins of Azimuthal Metallicity Variations in the Galactic Disk
*Advisor:* Keith Hawkins

**BS, Astrophysics**; University of California, Los Angeles (UCLA) 2024

## Research

**Graduate Student Researcher at UT Austin – Austin, TX** Sept. 2024 – Present
*Advisor:* Keith Hawkins

**Student Researcher at University of Notre Dame – Notre Dame, IN** May 2023 – Aug. 2024
*Advisor:* Lauren Weiss

*Combined radial velocity measurements of Kepler-34 and Kepler-35 from several spectrographs to place a strong mass upper limit on additional stellar and substellar companions in the two systems.*

**Student Researcher at UCLA – Los Angeles, CA** May 2022 – June 2024
*Advisor:* Smadar Naoz

*Utilized supercomputers to create population synthesis models and explore the spatial distribution of stellar remnants at the Galactic Center.*

## Publications

[1] **Jurado, C.**; Hawkins, K.; Hunt J. A. S.; Hackshaw, Z.; Filion C.; Frankel, N., Carr, C., "Dynamical Origins of Azimuthal Metallicity Variations in the Galactic Disk: Insights from kinematic ridges with Gaia", ApJ, 10.3847/1538-4357/ae346c

[2] **Jurado, C.**; Weiss, L. M.; Daclison L.; Tofflemire, B.M.; Orosz J. A.; Welsh, W. F., "Upper Limits on Stellar Companions to the Kepler-34 and Kepler-35 Systems", AJ, 10.3847/1538-3881/ada5f5

[3] **Jurado, C.**; Naoz, S.; Lam, C. Y.; Hoang, B.-M., "Natal Kicks from the Galactic Center and Implications on their Environment and the Roman Space Telescope," ApJ, doi.org/10.3847/1538-4357/ad55ee

## Conference Presentations

Division on Dynamical Astronomy Annual Meeting, Chicago, IL June 2026
*Contributing Talk:* "The Dynamical Origins of Azimuthal Metallicity Variations"

APS Far West Section, San Diego, CA — Oct. 2023
*Contributing Talk:* "The implications of Exploding Stars at the Galactic Center"

## Seminars & Informal Talks

Stars, Planets, and ISM Seminar, The University of Texas at Austin — Oct. 28 2026
*Talk:* "Detailed Chemical Abundances Of Hierarchical Triples"

Stars, Planets, and ISM Seminar, The University of Texas at Austin — April 8 2026
*Talk:* "Exploring the Chemo-Kinematic Structure of the Milky Way Disk"

Smadar Naoz Research Group Meeting, UCLA — Jan. 8 2026
*Talk:* "Dynamical Origins Of Azimuthal Metallicity Variations"

Stars, Planets, and ISM Seminar, The University of Texas at Austin — Jan. 29 2025
*Talk:* "The Effects of Natal Kicks in the Galactic Center"

AstroBash, UCLA Department of Physics and Astronomy — Oct. 18 2023

AstroBash, UCLA Department of Physics and Astronomy — Nov. 7 2022

## Teaching

Graduate Teaching Assistant, Stellar Astronomy (AST 352K) — Aug. 2026 – Dec. 2026
*Substitute Lecturer* (2 lectures, Nov. 2026)
"Kepler's Laws and Binary Star Systems"
"Galactic Kinematics and Stellar Populations"

Graduate Teaching Assistant, Search For Extraterrestrial Life (AST 309L) — Aug. 2025 – Dec. 2025

Guest Lecture, Introductory Astronomy (AST 307) — April 1, 2025
*Lecture*: "The Structure And Components Of The Milky Way Galaxy"

Graduate Teaching Assistant, Introductory Astronomy (AST 301) — Jan. 2025 – May 2025

Undergraduate Course Reader, Introductory Physics (Physics 5A) — Oct. 2023 – Dec. 2023

Undergraduate Course Reader, Introductory Mechanics (Physics 1A) — April 2024 – June 2024

Undergraduate Learning Assistant, Introductory Mechanics (Physics 1A) — Jan. 2023 – Mar. 2023

## Awarded Telescope Time

**Chemical Homogeneity of Hierarchical Stellar Triples** — Feb., July & Nov. 2026
*PI:* Carlos Jurado (15 nights)
2.7m Harlan J. Smith Telescope, McDonald Observatory

## Observing Experience

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| 2.7m Harlan J. Smith Telescope, McDonald Observatory (21 nights) | Ft Davis, TX |
| 0.8m Telescope, McDonald Observatory (2 nights) | Ft Davis, TX |

## Service & Outreach

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| Poster Presenter, Astronomy on Tap Austin 100th Show | July 2026 |
| Graduate Student Assembly Representative, The University of Texas at Austin | 2025 - 2026 |
| Student Panelist, UCLA AstroLive Summer Workshop | July 2022 |
| Volunteer and Student Panelist, UCLA Physics and Astronomy Outreach Event | May 2022 |

## Skills

*Programming Languages:* Python, SQL/ADQL, C/C++
*Tools:* iSpec, SAPHIRES, galpy
*Software:* BACCHUS, IRAF, SAOImageDS9
*Other:* Linux/Unix, Git, LaTeX